```
# ================================================================
# Kaggle Notebook Setup Cell
# ================================================================
%matplotlib inline
%load_ext rpy2
%load_ext cudf.pandas
%load_ext cuml.accel
%load_ext cuml.pandas
%load_ext cuml.numpy
%load_ext cuml.scipy
%load_ext cuml.sklearn

%pip install -q kaggle-datasets[gcsfs]*

# ================================================================
# Kaggle Dataset Download
# ================================================================
!kaggle competition download -c ieee-fraud-detection -p /kaggle/input/ieee-fraud-detection
!unzip -q -n /kaggle/input/ieee-fraud-detection/*.zip -d /kaggle/input/ieee-fraud-detection

# ================================================================
# Core Statistical & Data Tools
# ================================================================
%%R
install.packages('Matrix')
library('Matrix', lib.loc='/usr/local/lib/R/site-library')
library('data.table')
library('ggplot2')
library('scales')
library('Hmisc')
library('corrplot')

%pip install -q -U lightgbm catboost*

# ================================================================
# Environment Verification
# ================================================================
%%R
cat('[OK] Matrix', as.character(packageVersion('Matrix')), '\n')
cat('[OK] data.table', as.character(packageVersion('data.table')), '\n')
cat('[OK] ggplot2', as.character(packageVersion('ggplot2')), '\n')
cat('[OK] scales', as.character(packageVersion('scales')), '\n')
cat('[OK] Hmisc', as.character(packageVersion('Hmisc')), '\n')
cat('[OK] corrplot', as.character(packageVersion('corrplot')), '\n')

# ================================================================
# Core Statistical & Data Tools
# ================================================================
%%R
# ================================================================
# Core Data Manipulation
# ================================================================
library(dplyr)
library(tidyr)
library(stringr)
library(lubridate)

# ================================================================
# Extended Statistical Tools
# ================================================================
library(MASS)
library(Matrix)
library(boot)
library(Hmisc)

# ================================================================
# Machine Learning Tools
# ================================================================
library(caret)
library(randomForest)
library(glmnet)
library(xgboost)
library(lightgbm)
library(catboost)

# ================================================================
# Visualization Tools
# ================================================================
library(ggplot2)
library(scales)
library(corrplot)
library(pheatmap)
library(ggrepel)

# ================================================================
# Utilities
# ================================================================
library(jsonlite)
library(yaml)
library(rmarkdown)

# ================================================================
# Verification
# ================================================================
cat('[OK] dplyr', as.character(packageVersion('dplyr')), '\n')
cat('[OK] tidyr', as.character(packageVersion('tidyr')), '\n')
cat('[OK] stringr', as.character(packageVersion('stringr')), '\n')
cat('[OK] lubridate', as.character(packageVersion('lubridate')), '\n')
cat('[OK] MASS', as.character(packageVersion('MASS')), '\n')
cat('[OK] Matrix', as.character(packageVersion('Matrix')), '\n')
cat('[OK] boot', as.character(packageVersion('boot')), '\n')
cat('[OK] Hmisc', as.character(packageVersion('Hmisc')), '\n')
cat('[OK] caret', as.character(packageVersion('caret')), '\n')
cat('[OK] randomForest', as.character(packageVersion('randomForest')), '\n')
cat('[OK] glmnet', as.character(packageVersion('glmnet')), '\n')
cat('[OK] xgboost', as.character(packageVersion('xgboost')), '\n')
cat('[OK] lightgbm', as.character(packageVersion('lightgbm')), '\n')
cat('[OK] catboost', as.character(packageVersion('catboost')), '\n')
cat('[OK] ggplot2', as.character(packageVersion('ggplot2')), '\n')
cat('[OK] scales', as.character(packageVersion('scales')), '\n')
cat('[OK] corrplot', as.character(packageVersion('corrplot')), '\n')
cat('[OK] pheatmap', as.character(packageVersion('pheatmap')), '\n')
cat('[OK] ggrepel', as.character(packageVersion('ggrepel')), '\n')
cat('[OK] jsonlite', as.character(packageVersion('jsonlite')), '\n')
cat('[OK] yaml', as.character(packageVersion('yaml')), '\n')
cat('[OK] rmarkdown', as.character(packageVersion('rmarkdown')), '\n')

# ================================================================
# Data Loading
# ================================================================
train <- read.csv('/kaggle/input/ieee-fraud-detection/train_transaction.csv')
cat('train dim:', dim(train), '\n')
cat('train size:', format(object.size(train), units='MB'), '\n')

# ================================================================
# Basic Exploration
# ================================================================
dim(train)
nrow(train)
ncol(train)
head(train, 5)
tail(train, 5)
str(train)
summary(train)

# ================================================================
# Target Variable Analysis
# ================================================================
table(train$isFraud)
prop.table(table(train$isFraud))
cat('Fraud rate:', mean(train$isFraud), '\n')

# ================================================================
# Missing Value Analysis
# ================================================================
colSums(is.na(train))
colSums(is.na(train))/nrow(train)
sort(colSums(is.na(train))/nrow(train), decreasing=TRUE)[1:20]

# ================================================================
# Numeric Variable Analysis
# ================================================================
numeric_cols <- sapply(train, is.numeric)
sum(numeric_cols)
colnames(train)[numeric_cols]
summary(train[, c('TransactionAmt', 'TransactionDT')])

# ================================================================
# Categorical Variable Analysis
# ================================================================
cat_cols <- colnames(train)[!numeric_cols]
length(cat_cols)
cat_cols
sapply(train[, cat_cols], function(x) length(unique(x)))

# ================================================================
# Time Analysis
# ================================================================
hist(train$TransactionDT, breaks=100, main='TransactionDT')
hist(train$TransactionAmt, breaks=100, main='TransactionAmt')
hist(log10(train$TransactionAmt+1), breaks=100, main='log10(TransactionAmt)')

# ================================================================
# Correlation Analysis
# ================================================================
cor_mat <- cor(train[, c('TransactionAmt', 'TransactionDT')])
cor_mat

# ================================================================
# Save Processed Data
# ================================================================
saveRDS(train, '/kaggle/working/train.rds')
cat('saved train.rds\n')

# ================================================================
# Load Identity Data
# ================================================================
train_id <- read.csv('/kaggle/input/ieee-fraud-detection/train_identity.csv')
cat('train_id dim:', dim(train_id), '\n')
cat('train_id size:', format(object.size(train_id), units='MB'), '\n')

# ================================================================
# Basic Exploration
# ================================================================
dim(train_id)
nrow(train_id)
ncol(train_id)
head(train_id, 5)
tail(train_id, 5)
str(train_id)
summary(train_id)

# ================================================================
# Missing Value Analysis
# ================================================================
colSums(is.na(train_id))
colSums(is.na(train_id))/nrow(train_id)
sort(colSums(is.na(train_id))/nrow(train_id), decreasing=TRUE)[1:20]

# ================================================================
# Numeric Variable Analysis
# ================================================================
numeric_cols <- sapply(train_id, is.numeric)
sum(numeric_cols)
colnames(train_id)[numeric_cols]
summary(train_id[, c('id_01', 'id_02', 'id_03', 'id_04', 'id_05', 'id_06', 'id_07', 'id_08', 'id_09', 'id_10', 'id_11')])

# ================================================================
# Categorical Variable Analysis
# ================================================================
cat_cols <- colnames(train_id)[!numeric_cols]
length(cat_cols)
cat_cols
sapply(train_id[, cat_cols], function(x) length(unique(x)))

# ================================================================
# Save Processed Data
# ================================================================
saveRDS(train_id, '/kaggle/working/train_id.rds')
cat('saved train_id.rds\n')

# ================================================================
# Merge Transaction and Identity
# ================================================================
train_merged <- merge(train, train_id, by='TransactionID', all.x=TRUE)
cat('merged dim:', dim(train_merged), '\n')
cat('merged size:', format(object.size(train_merged), units='MB'), '\n')

# ================================================================
# Basic Exploration
# ================================================================
dim(train_merged)
nrow(train_merged)
ncol(train_merged)
head(train_merged, 5)
tail(train_merged, 5)
str(train_merged)
summary(train_merged)

# ================================================================
# Missing Value Analysis
# ================================================================
colSums(is.na(train_merged))
colSums(is.na(train_merged))/nrow(train_merged)
sort(colSums(is.na(train_merged))/nrow(train_merged), decreasing=TRUE)[1:20]

# ================================================================
# Numeric Variable Analysis
# ================================================================
numeric_cols <- sapply(train_merged, is.numeric)
sum(numeric_cols)
colnames(train_merged)[numeric_cols]
summary(train_merged[, c('id_01', 'id_02', 'id_03', 'id_04', 'id_05', 'id_06', 'id_07', 'id_08', 'id_09', 'id_10', 'id_11')])

# ================================================================
# Categorical Variable Analysis
# ================================================================
cat_cols <- colnames(train_merged)[!numeric_cols]
length(cat_cols)
cat_cols
sapply(train_merged[, cat_cols], function(x) length(unique(x)))

# ================================================================
# Save Processed Data
# ================================================================
saveRDS(train_merged, '/kaggle/working/train_merged.rds')
cat('saved train_merged.rds\n')

# ================================================================
# Load Merged Data
# ================================================================
train <- readRDS('/kaggle/working/train_merged.rds
```

```
import pulsar.schema.*;

// AutoGeneratedSchemaDef(version="1.0", propertyMap={})
public class ClientMessageRedeliveryCount implements SchemaDefinition<ClientMessageRedeliveryCount> {
    // AutoGeneratedSchemaDef(version="1.0", propertyMap={})
    public int messageId, redeliveryCount;
    // AutoGeneratedSchemaDef(version="1.0", propertyMap={})
    public long[] ackSets;
}
```